

YUANKAI TAO

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EDUCATION

Georgia Institute of Technology <i>M.S. in Operations Research, Stewart School of Industrial and Systems Engineering</i> Selected Coursework: Deterministic Optimization, Stochastic Process, Simulation	<i>Aug 2026 — Dec 2027</i>
University of California, San Diego <i>M.S. in Quantitative Finance, Rady School of Management</i> Selected Coursework: Deep Learning, Computational Finance Methods, Data Science for Finance	<i>Sep 2024 — Dec 2025</i> GPA: 3.84/4.0
Zhongnan University of Economics and Law <i>B.S. in Management, School of Accounting and Finance</i> Selected Coursework: Probability & Statistics, Linear Algebra, Calculus, Python Programming	<i>Sep 2020 — Jun 2024</i> GPA: 3.63/4.0

TECHNICAL SKILLS

Programming & Tools	Python, SQL, MATLAB, MS Office, L ^A T _E X, Git, Linux, GCP
Quantitative Methods	Probability Theory, Simulation, Stochastic Process, Optimization, ML & DL
Libraries & Frameworks	Pandas, Polars, Numpy, Scipy, PyTorch, scikit-learn, CVXPY, Backtrader

PROFESSIONAL EXPERIENCE

Guangdong Hedge Fund Investment Management Co., Ltd. Quantitative Research, Intern	<i>May 2026 — Jul 2026</i> <i>Quantitative Strategy Department Guangdong, China</i>
- Engineered end-to-end commodity futures ML pipeline with 16 features (trend, momentum, volatility, liquidity, volume, open-interest); trained and ensemble LightGBM, XGBoost, CatBoost, and neural networks across 1- to 20-day horizons using rolling out-of-sample validation with both statistical and trading metrics. - Developed a monthly rebalanced cross-sectional futures strategy using LightGBM return probabilities with 120-day rolling retraining window; implemented in Backtrader achieving 1.89 Sharpe ratio after transaction costs, 16.79% annualized return, and 25.51% excess return. - Built a dynamic VaR/CVaR risk measurement system for metal futures; constructed 17 risk models combining 5 volatility models (GARCH, EWMAV) with 3 distribution assumptions (Normal, Student-t, Skewed-t, Filtered Historical Simulation); introduced semi-variance slicing method to isolate long/short momentum risks.	
VariFlight Technology Data Analyst, Intern	<i>Jul 2025 — Sep 2025</i> <i>Data Development Department Anhui, China</i>
- Utilized a Python-based stack for web scraping to gather public reports and web pages/images from the Civil Aviation Administration of China (CAAC). - Employed both traditional and deep learning-based OCR/image recognition tools including pytesseract, Camelot, etc. to extract unstructured data from 600+ non-formatted pdf reports and 500+ webpages and automated the transfer from raw data into formatted csv files as input for further local LLM training.	
Ernst & Young Assurance, Intern	<i>Dec 2022 — Feb 2023</i> <i>Financial Service Office (FSO)</i>
Assisted the client, Hyundai Motor Finance Company, in auditing financial data related to inter-company and investment transactions; Conducted variance analysis to investigated significant deviations.	

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill <i>The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining</i>	<i>Mar 2025 — Jan 2026</i>
Engineered large-scale data pipeline (Python, PostgreSQL, PostGIS) processing 76-million-observation spatial dataset on Google Cloud Platform; managed 964.5M-record SQL database with 100GB+ data on UNC Longleaf HPC cluster using Python and DuckDB.	

- Implemented parallelized spatial computation workflows to calculate comparable housing variables across multiple distance ranges with property-specific filtering criteria, achieving approximately $50\times$ computational speedup.
- Executed complex geospatial queries to extract population statistics with time-window and property-related constraints.

SELECTED PROJECTS

Kaggle Jane Street Market Prediction Competition

Silver Medal

- Processed 47M minute-level observations spanning 1,669 trading days; engineered cross-sectional, lagged, rolling, and market-regime features from 79 anonymized variables and 9 responder targets.
- Trained and ensembled LightGBM, CatBoost, XGBoost, MLP, and TabNet models using time-aware validation; achieved 94th place (Top 2.5%) of 3,757 teams on private leaderboard.

UCSD / FMSbonds, Inc. 2025 Capstone Project

Aug 2025 — Dec 2025

Team Lead | High-Performance Bond Pricing Engine Optimization

- Completed the self-defined library by developing a comprehensive municipal bond risk-screening framework covering credit, structural, sector, tax, interest-rate and data integrity risks.
- Designed caching system for high-frequency cash flow queries, eliminating redundant cash flow object generation; reduced per-bond solving time from 1.17 ms to 0.13 ms ($9\times$ speedup).
- Rewrote yield-to-worst numerical solver using Numba JIT compilation, reducing computation time from 1.57 ms to $7.5\ \mu\text{s}$ per bond ($210\times$ speedup).

CERTIFICATIONS & HONORS

Deep Learning Specialization (5-Course Certificate)

Coursera, DeepLearning.AI

Algorithms and Data Structures Certificate (6-Course Certificate)

Coursera, UC San Diego

Beta Gamma Sigma (Top 20% of graduating class)

UCSD, 2024 — 2025

WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%)

2024

Special Scholarship in Innovation and Entrepreneurship

ZUEL, 2022 — 2023